Marked-Root Keller Maps and Degreewise Stable Multiplicity

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Abstract

We place two families of polynomial Keller maps of affine three-space in a common marked-root framework. For every generic degree $N \geq 4$, a split weighted map and all parameter branches of one cancellation type for each proper divisor of $N-1$ are constructed. We compute the complete boundary of the canonical finite normalization, prove functoriality under stable polynomial left–right equivalence, and compare the scheme-theoretic contact of two intrinsically marked target divisors. We also retain the full Fitting divisor of the discriminant normalization, its off-diagonal node-pairing scheme, and its conductor. On an explicitly nonempty ordinary boundary-clean open, the decorated-normalization invariant is etale of degree $N-2$ onto the normalization of its reduced image, which has dimension $N-3$; hence weighted degree-N maps contain an $(N-3)$-dimensional family of stable classes for every $N \geq 4$. More generally, for every normalized admissible seed whose zero root has multiplicity at least two, the richer full marked incidence cover with its regular-reconstruction open is faithful, even when the other roots collide. The marked admissible-cover compactification has the same normalized Stein factors and conductor square at arbitrary simultaneous root collisions. Its marked-zero-fiber LL morphism has degree $(N-2)N^{N-3}$. Separately, the weighted contact is reduced, whereas a cancellation type (m, r) has discriminant ramification $r+1$ and nilpotency index $mr(m+1)$. For fixed (m, r), the full marked normalization with its affine reconstruction open also recovers the selected root of the degree-mr cancellation polynomial. Thus all parameter roots give distinct stable classes. Writing σ for the sum-of-divisors function, the construction therefore distinguishes at least $1 + (N-1)\tau(N-1) - \sigma(N-1)$ stable classes of degree N, each with a nontrivial fiber.

1 Introduction

Within the accompanying repository, this paper is the canonical statement and proof of the degreewise stable-multiplicity theorem. Other repository documents may summarize the result or audit individual inputs, but should defer to the formulation here.

A polynomial map is called Keller when its Jacobian determinant is a nonzero constant. Recent explicit three-dimensional examples are most naturally understood by marking a root of a finite inverse equation. The original cubic example, its projective marked-root interpretation, its exact image theorem, and the subsequent weighted construction motivate the language below, but none is used here as a black-box dependency. We give the formulas and the parts of their proofs needed for the degreewise result.

The second family is produced by finite cancellation of the apparent poles in an integral inverse coordinate. Although the source formulas look very different, both families have the same inverse skeleton: a finite root cover, rational reconstruction, and an affine source equal to the open on which reconstruction is regular. Missing root sheets then become divisors in the canonical finite normalization.

For provenance, the initial explicit map was publicly attributed to Levent Alpöge and Fable; Andy Jiang described the marked-projective-root organization, and Alexis Gallagher developed the weighted-lift viewpoint. Those discoveries motivate the present setup. The cancellation construction, canonical boundary comparison, and degreewise count are the later results being isolated here. This wording records the currently located public trail and is not a priority claim.

The positive-dimensional weighted-locus result is Corollary 8.13 below. The following theorem is the complementary discrete cross-construction statement: it distinguishes the weighted locus from every cancellation parameter branch, separates the cancellation types, and separates the parameter roots within each type.

Theorem 1.1. *For every integer $N \geq 4$, there exist at least*

$$1 + (N-1)\tau(N-1) - \sigma(N-1)$$

pairwise stably polynomially left–right inequivalent maps, where σ denotes the sum-of-divisors function,

$$F : \mathbb{A}^3_{\mathbb{C}} \longrightarrow \mathbb{A}^3_{\mathbb{C}}$$

with nonzero constant Jacobian, generic fiber degree N, and a fiber containing N distinct points.

The count comes from one weighted class and all $N-1-r$ parameter roots of the cancellation type belonging to each proper divisor $r \mid N-1$. Five lemmas separate the construction, exhaustion, invariance, and intersection calculations; the parameter-faithfulness theorem then separates roots within a type. This separation is important: a boundary component visible in a resolvent chart is not yet a canonical invariant until all boundary components have been found.

2 Marked roots and canonical boundary

Let Y be normal affine and let $F : U \to Y$ be dominant and quasi-finite, with U normal. Put $K = k(Y)$, $L = k(U)$, and

$$\overline{X}_F = \mathrm{Norm}_Y(L).$$

The normalization is finite over Y. Zariski's Main Theorem identifies U with a distinguished open in $\overline{X}_F$, and we put

$$\partial_F = (\overline{X}_F \setminus U)_{\mathrm{red}}.$$

A *marked-root presentation* consists of a degree-N separable equation $\Psi(T; y) = 0$ over K, its normalized incidence over Y, and rational reconstruction functions whose common regular-reconstruction open is $U \simeq \mathbb{A}^3$. The equation and primitive element are auxiliary; the pair $(\overline{X}_F, U)$ is canonical.

For a boundary prime E mapping to a target divisor Z, retain its ramification and residue degrees $(e(E/Z), f(E/Z))$. We also retain reduced and scheme-theoretic multiple intersections of the target boundary-image divisors. In the families below exactly two target vertices occur. The first is intrinsically marked as the unique one receiving a ramified boundary prime and is denoted Z_Δ; the other is Z_0. Define

$$e_\Delta = e(E_\Delta/Z_\Delta), \qquad \mu = \min\{a \geq 1 : \mathrm{Nil}(\mathcal{O}_{Z_\Delta \cap Z_0})^a = 0\}. \tag{2.1}$$

Here the scheme intersection is defined by the sum of the two reduced prime divisor ideals. Thus $\mu = 1$ when the intersection is reduced.

3 The two constructions

3.1 Weighted marked roots

Let $H \in k[W]$ satisfy

$$H(0) = H'(0) = H(1) = 0, \qquad H'(1) = -c \neq 0, \qquad \kappa = H''(1)/c \neq -2. \tag{3.1}$$

Choose $b_0 \neq 0$ and $a_0 = -(1+\kappa)/(2+\kappa)$. Set

$$v = xy, \quad S = x^2 z, \quad u = 1 + v, \quad \gamma = 1 + a_0 v + b_0 S, \quad W = u\gamma,$$

and, with $p = H'$ and $q = (Wp - H)/c$, define

$$C = x\gamma, \qquad B = \frac{c + p(W)/\gamma}{x}, \qquad A = \frac{u + q(W)/\gamma^2}{x^2}. \tag{3.2}$$

The conditions (3.1) make the displayed quotients polynomial. Expanding at $x = 0$ shows that the constant and linear terms vanish; the remaining linear obstruction is exactly $(2+\kappa)a_0 + (1+\kappa) = 0$. Direct triangular differentiation gives

$$\det \frac{\partial(A, B, C)}{\partial(x, y, z)} = b_0 c. \tag{3.3}$$

The marked inverse equation is

$$H(W) - BCW + cAC^2 = 0. \tag{3.4}$$

At every generic simple root, differentiating (3.4) and using (3.2) reconstructs (x, y, z); hence its degree is the generic degree of the map.

Proposition 3.1 (explicit regular weighted fiber)**.** *Suppose $H \in \overline{\mathbb{Q}}[W]$ is an admissible seed of degree d and $c \in \overline{\mathbb{Q}}^{\times}$. For any complex number α transcendental over $\overline{\mathbb{Q}}$, the target*

$$(A, B, C) = (\alpha, 0, 1) \tag{3.5}$$

has exactly d distinct inverse roots, and every one lies in the regular-reconstruction open. More explicitly, if

$$H(w_i) + c\alpha = 0, qquad \gamma_i = -\frac{H'(w_i)}{c}, qquad 1 \leq i \leq d, \tag{3.6}$$

then $H'(w_i) \neq 0$ and the d source points are

$$x_i = \gamma_i^{-1}, \qquad y_i = w_i - \gamma_i,$$

$$z_i = \frac{\gamma_i^2}{b_0}\left[\gamma_i - 1 - a_0\left(\frac{w_i}{\gamma_i} - 1\right)\right]. \tag{3.7}$$

Proof. At (3.5), the inverse equation and its W-derivative are

$$E(W) = H(W) + c\alpha, qquad E_W(W) = H'(W). \tag{3.8}$$

If w were a common root, then w would be algebraic over $\overline{\mathbb{Q}}$ because it is a root of the nonzero polynomial H'. Hence $H(w)$ would be algebraic, contradicting $H(w) = -c\alpha$. Thus E is a degree-d polynomial with d distinct complex roots.

The reconstruction formulas are

$$\gamma = -E_W/c, \qquad x = C/\gamma, qquad y = (W-\gamma)/C,$$
$$z = \frac{\gamma - 1 - a_0(W/\gamma - 1)}{b_0 x^2}. \tag{3.9}$$

Their only possible denominators on the incidence are C, E_W (through γ), and the fixed units b_0, c. At (3.5), $C = 1$ and the preceding coprimality argument gives $E_W(w_i) \neq 0$ for every root. Substitution into (3.9) gives (3.7), so all reconstruction denominators are excluded simultaneously. Distinct w_i give distinct marked source points. $\square$

Lemma 3.2 (weighted admissibility)**.** *For every $N \geq 4$,*

$$H_N(W) = W^2(1-W)(1+W^{N-3}) \tag{3.10}$$

is an admissible split seed of degree N. With $b_0 = 1$, (3.2) is a Keller map of generic degree N, and the fiber over $(\pi, 0, 1)$ consists of N distinct points.

Proof. The roots of $1 + W^{N-3}$ are distinct in characteristic zero, nonzero, and different from 1. Direct differentiation gives

$$H_N'(1) = -2, qquad H_N''(1)/2 = -(N+1) \neq -2.$$

Thus (3.1) holds with $c = 2$. Equations (3.2)–(3.4) give polynomiality, Jacobian 2, and exact generic degree N. Since $H_N \in \mathbb{Q}[W]$ and π is transcendental, Proposition 3.1 applies with $\alpha = \pi$ and supplies the asserted N-point fiber, with all reconstruction denominators nonzero. $\square$

3.2 Cancellation marked roots

Fix $m, r \geq 1$, $C \in k^*$, and a polynomial $h(A)$. Put

$$A = 1 + xy^m, \quad B = A^{r+1}z + y^{m+1}h(A), \quad P = AB, \quad Q = y + xB, \tag{3.11}$$

and initially in the localization at A put

$$R = C\int_0^{x/A} \{1 - t(Q - Pt)^m\}^r \, dt. \tag{3.12}$$

Writing $s = x/A$, one has

$$y = Q - sP, \quad D = 1 - s(Q - sP)^m = A^{-1}, \quad x = s/D,$$

and the change $(x, y, z) \mapsto (s, P, Q)$ has determinant $-A^r$. Since $R_s = CD^r = CA^{-r}$,

$$\det \frac{\partial(P, Q, R)}{\partial(x, y, z)} = -C. \tag{3.13}$$

Only the z-free part of (3.12) can have an A-pole. Its regularity is the finite condition

$$\Phi_{m,r}(A, h(A)) \equiv 0 \pmod{A^{r+1}}, \tag{3.14}$$

where

$$\Phi_{m,r}(A, H) = \int_0^1 [A + (1-A)u\{1 - (1-A)H(1-u)\}^m]^r \, du.$$

If $q = h(0)$, the constant term of (3.14), after monic normalization, is

$$M_{m,r}(q) = \sum_{j=0}^{mr} (-1)^j \binom{mr + r + 1}{j} q^{mr-j}. \tag{3.15}$$

Lemma 3.3 (divisor construction)**.** *Let $N \geq 4$, put $n = N - 1$, and let r be a proper positive divisor of n. For $m = n/r - 1$, the polynomial $M_{m,r}$ has $mr = N - 1 - r$ distinct complex roots. Every root determines a unique normalized polynomial h satisfying (3.14), and equations (3.11)–(3.12) define a polynomial Keller map of generic degree N with a fiber of N distinct points.*

Proof. We have $m \geq 1$. Up to a nonzero scalar, (3.15) is

$$y(q) = \int_0^1 u^r(1 - q + qu)^{mr}\, du = \frac{1}{r+1}{}_2F_1(-mr, 1; r+2; q).$$

Its differential equation is

$$q(1-q)y'' + [r + 2 - (2 - mr)q]y' + mr\, y = 0. \tag{3.16}$$

If y and y' vanished together away from $0, 1$, uniqueness in (3.16) would make y identically zero. At 0 and 1 its integral is nonzero. Thus $M_{m,r}$ is squarefree. It has degree $mr \geq 1$, hence has a complex root q. At such a root, $\partial_q \Phi_{m,r}(0, q) \neq 0$; successively equating the coefficients of $A, \ldots, A^r$ determines a unique jet $h = q + h_1 A + \cdots + h_r A^r$ satisfying (3.14). Thus R is polynomial and (3.13) applies.

The inverse equation is

$$\Psi(T) = C \int_0^T \{1 - t(Q - Pt)^m\}^r\, dt - R, \tag{3.17}$$

of degree $r(m+1) + 1 = N$. If $D = 1 - T(Q - PT)^m \neq 0$, reconstruction is

$$y = Q - TP, \quad A = D^{-1}, \quad x = TD^{-1},$$
$$z = PD^{r+2} - y^{m+1}D^{r+1}h(D^{-1}). \tag{3.18}$$

The resultant of Ψ and D is generically nonzero, so (3.18) proves exact generic degree. At $(P, Q, R) = (1, 0, 0)$, (3.17) is $CTE_{m,r}((-1)^m T^{m+1})$. A repeated nonzero root would force $(-1)^m T^{m+1} = 1$, but $E_{m,r}(1) = \int_0^1 (1 - v^{m+1})^r\, dv \neq 0$. Hence all N roots are simple and reconstruct to N distinct source points. □

4 Exhausting the normalization boundary

This section gives the codimension-one calculation on which the rest of the paper depends. A geometric branch over Z means a prime after passing from the generic DVR of Z to its strict henselization; it has residue degree one. The tables below are instead written over the displayed generic DVR whenever a moving residual polynomial is more naturally kept irreducible. After strict henselization such a row splits into its geometric branches. In general the arithmetic residue degree is the degree of the corresponding irreducible residual factor.

Let $Z \subset Y$ be a prime divisor, let $\mathcal{O}_{Y,\eta_Z}$ have uniformizer π_Z, and let S_Z be its integral closure in $L = k(U)$. Normality makes S_Z a finite semilocal Dedekind domain. If $[L : K] = N$, then

$$N = \dim_{k(Z)} S_Z/\pi_Z S_Z = \sum_{E|Z} e(E/Z)f(E/Z). \tag{4.1}$$

Every summand is positive and the sum includes primes whose generic points belong to the affine-source open. Thus a displayed list which reaches N is the complete list over Z.

We use reduced discriminant equations. The critical-value maps below have integral source and height-one prime kernel in the target UFD. We denote a generator of that kernel by δ and put $Z_\Delta = V(\delta)$. This is important for cancellation maps: their raw polynomial discriminant has generic multiplicity r, whereas δ occurs once.

Lemma 4.1 (exact weighted discriminant saturation)**.** *For*

$$H_N(W) = W^2(1-W)(1+W^{N-3}), \qquad E_N(W) = H_N(W) - BCW + 2AC^2,$$

there are nonzero constants $\lambda_N, \mu_N \in \mathbb{C}^\times$ and an irreducible polynomial $\delta_N \in \mathbb{C}[A,B,C]$ such that

$$\mathrm{Disc}_W(E_N) = \lambda_N C^2 \delta_N, \qquad \delta_N|_{C=0} = \mu_N(B^2 - 8A). \tag{4.1a}$$

In particular, $C \nmid \delta_N$, the residual discriminant $V(\delta_N)$ is prime and reduced, and its generic normalization prime is a boundary prime with $(e,f) = (2,1)$.

Proof. Work first over the generic DVR $\mathbb{C}(A,B)[[C]]$. The zero of H_N at the origin has exact multiplicity two and leading coefficient one. In the scaled chart $W = Cu$,

$$C^{-2}E_N(Cu) = u^2 - Bu + 2A + O(C). \tag{4.1b}$$

Thus the two roots specializing to zero differ by C times a unit over the generic quadratic extension, and the square of their difference contributes exactly $C^2(B^2-8A)$ to the root-product formula for the discriminant. Every other root specializes to one of the distinct nonzero simple roots of H_N; all its differences from the two small roots and from the other simple roots are units. Consequently

$$v_C(\mathrm{Disc}_W E_N) = 2, \quad C^{-2}\,\mathrm{Disc}_W(E_N)\big|_{C=0} = \nu_N(B^2 - 8A)$$

for some $\nu_N \in \mathbb{C}^\times$. This proves that the forced power is exactly C^2 and that the quotient is not divisible by C.

It remains to identify that quotient intrinsically. On $C \neq 0$ the critical incidence is

$$B = \frac{H_N'(W)}{C}, \qquad A = \frac{WH_N'(W) - H_N(W)}{2C^2}.$$

It is integral and generically birational to its image: for $s = H_N'(W)$ and $t = WH_N'(W) - H_N(W)$ one has $dt/ds = W$ wherever $H_N''(W) \neq 0$. Its height-one kernel is therefore generated by an irreducible polynomial δ_N. The generic critical root is exactly double, so the discriminant vanishes to order one along this image. Hence the C-primitive quotient of the discriminant is a nonzero scalar multiple of δ_N, proving (4.1a), primeness, and reducedness.

At the generic point of $V(\delta_N)$, C is a unit. If $u = E_{N,W}$ is the uniformizer of the normalized critical incidence, then

$$\delta_N = \varepsilon u^2, \qquad \varepsilon \in \widehat{\mathcal{O}}^\times_{\overline{X},E_\Delta}.$$

The critical-value map is birational, so $f = 1$, and the displayed equation gives $e = 2$. This prime cannot meet the affine-source open: the Keller map is etale there, whereas this DVR extension is ramified. It is therefore exactly the claimed ramified boundary vertex. $\square$

Lemma 4.2 (boundary exhaustion)**.** *The complete boundary-image list for the split weighted map is $\Delta_H = V(\delta_H)$ and $C = 0$. Over Δ_H there is one boundary prime with $(e,f) = (2,1)$; over $C = 0$ there are $N-3$ geometric boundary primes with $(e,f) = (1,1)$.*

For the cancellation map the complete list is $\Delta_{m,r} = V(\delta_{m,r})$ and $P = 0$. Over the first there is one boundary prime with $(e,f) = (r+1,1)$; over the second there are $mr-1$ geometric boundary primes with $(e,f) = (1,1)$. For $N \geq 4$ both target vertices occur. In each family the discriminant is the unique vertex receiving ramification.

Proof. We give the local calculation family by family.

Weighted discriminant. For the degreewise seed H_N, Lemma 4.1 already proves the exact C^2 saturation and identifies the reduced prime equation. The following local description records the corresponding generic DVR and also applies to any split seed satisfying the same simple-root hypotheses. Write $E(W) = H(W) - BCW + cAC^2$. On $C \neq 0$, the critical incidence is

$$E = E_W = 0, \quad B = \frac{H'(W)}{C}, \quad A = \frac{WH'(W) - H(W)}{cC^2}. \tag{4.2}$$

It is parameterized by $(W, C) \in \mathbb{A}^1 \times \mathbb{G}_m$ and is integral. It is generically birational: if $s = H'(W)$ and $t = WH'(W) - H(W)$, then, where $H''(W) \neq 0$, one has $dt/ds = W$. Thus $W \in \mathbb{C}(s,t)$, while C is already the third target coordinate. Hence its height-one kernel $(\delta_H) \subset \mathbb{C}[A,B,C]$ is prime. Consequently $\Delta_H = V(\delta_H)$ is irreducible and reduced.

At its generic point $H''(W) \neq 0$ and exactly one inverse root is double. In the normalized local ring put $u = E_W = H'(W) - BC$. The critical divisor is $u = 0$, so u is an upstairs uniformizer. Taylor expansion in the root coordinate gives

$$\delta_H = \varepsilon u^2, \quad \varepsilon \in \widehat{\mathcal{O}}^{\times}_{\overline{X}, E_\Delta}. \tag{4.3}$$

Thus the generic DVR extension has $(e,f) = (2,1)$. The ordinary polynomial discriminant is a unit times δ_H at this DVR, not a higher power, because the generic double-root discriminant is transverse. Globally it may also contain a power of C; equivalently, the primitive part of $(\operatorname{Disc}_W E) : C^\infty$ contains δ_H once. The divisor $C = 0$ is audited separately below. The critical prime is outside the affine source: $E_W = -c\gamma$ makes reconstruction force $\gamma = 0$, and a ramified prime cannot lie in the etale Keller open. After strict henselization the other $N-2$ roots are simple affine branches; their local uniformizer is δ_H and each has $(e,f) = (1,1)$. Before strict henselization, each irreducible factor of the residual simple-root quotient gives one affine prime with $e = 1$ and f equal to that factor's degree; the degrees sum to $N - 2$. Hence

$$2 + (N-2) = N. \tag{4.4}$$

Weighted divisor $C = 0$. The ideal (C) is prime and reduced. Its generic DVR has uniformizer C and residue field $\mathbb{C}(A,B)$. Since $E(W)|_{C=0} = H(W)$, its complete local branch table is

root chart	generic local equation	(e,f)	position
$W = Cu$	$h_2u^2 - Bu + cA + O(C) = 0$	$(1,2)$	affine
$W = 1 + O(C)$	$E_W(1) = -c$	$(1,1)$	affine
$W = \rho_j + O(C)$	$E_W(\rho_j) = H'(\rho_j) \neq 0$	$(1,1)$	boundary.

(4.5)

In every row C remains an upstairs uniformizer. The first residual quadratic is irreducible over $\mathbb{C}(A,B)$ and generically separable, giving $f = 2$. Direct substitution in reconstruction gives nonnegative valuation for every source coordinate in the first two rows. In the last row,

$$y = \frac{W - \gamma}{C}, \quad (W-\gamma)|_{C=0} = \rho_j + \frac{H'(\rho_j)}{c} \neq 0, \tag{4.6}$$

so y has valuation -1. For H_N the ρ_j are the $N-3$ distinct roots of $1 + W^{N-3}$. The displayed numerator is nonzero for these roots when $N \geq 5$. In the sole exceptional case $N = 4$, $\rho = -1$ makes its constant term vanish, but the next normalized jet is

$$W = -1 - BC + \frac{3B^2 - A}{2}C^2 + O(C^3), \quad C^2 z \longrightarrow 2, \tag{4.7}$$

so that branch is still boundary. The degree sum is

$$2 + 1 + (N - 3) = N. \tag{4.8}$$

Cancellation discriminant. Put $D(T) = 1 - T(Q - PT)^m$. On $P \neq 0$ the inverse polynomial is monic up to a unit and $\Psi_T = CD(T)^r$. With $Y_0 = Q - PT$, the critical incidence has

$$T = Y_0^{-m}, \quad P = (Q - Y_0)Y_0^m, \tag{4.9}$$

and R equal to the critical value. It is integral, and generic critical values are distinct. To see the latter without a genericity assertion, specialize $Q = 0$: the $m+1$ critical roots differ by $(m+1)$-st roots of unity and their critical values are those roots times the same nonzero beta integral $C\int_0^1 (1 - s^{m+1})^r \, ds$. Thus a target point on a dense open of the discriminant has a unique critical preimage, so the critical-value map is birational to its image. Its height-one kernel $(\delta_{m,r}) \subset \mathbb{C}[P, Q, R]$ is prime and defines the reduced irreducible divisor $\Delta_{m,r}$.

At the generic critical point $u = D(T)$ is an upstairs uniformizer. Since D has a simple zero in the root direction, integration gives

$$\delta_{m,r} = \varepsilon u^{r+1}, \quad \varepsilon \in \widehat{\mathcal{O}}^{\times}_{\overline{X}, E_\Delta}. \tag{4.10}$$

Hence the critical boundary prime has $(e, f) = (r + 1, 1)$. The raw polynomial discriminant is generically a unit times $\delta^r_{m,r}$; further powers of P arise only from clearing the leading coefficient of the T chart. The reduced equation used here is the primitive generator of

$$\sqrt{(\mathrm{Res}_T(\Psi, D)) : P^\infty} = (\delta_{m,r}), \tag{4.11}$$

not the scheme discriminant. The ramified prime is outside the affine source because $A = D^{-1}$ has negative valuation. All other geometric roots are simple affine branches with $(e, f) = (1, 1)$ after strict henselization. Before that base change, the irreducible factors of the residual simple-root quotient give all affine primes, with $e = 1$, residue degrees equal to the factor degrees, and total residue degree $N - (r + 1)$. Thus

$$(r + 1) + N - (r + 1) = N. \tag{4.12}$$

Cancellation divisor $P = 0$. The ideal (P) is prime and reduced. Since the T chart is not monic there, put $U = PT$ and

$$J(U, P) = \int_0^U \{P - V(Q - V)^m\}^r \, dV.$$

The integral degree-N chart

$$CJ(U, P) - RP^{r+1} = 0 \tag{4.13}$$

has the source function field: it is primitive and linear in R, and $T = U/P$ over $P \neq 0$. At $P = 0$ it factors as

$$J(U, 0) = (-1)^r U^{r+1} Q^{mr} K_{m,r}(U/Q), \tag{4.14}$$

where

$$K_{m,r}(w) = w^{-(r+1)} \int_0^w v^r (1 - v)^{mr} \, dv. \tag{4.15}$$

The polynomial $K_{m,r}$ has degree mr and is squarefree. A nonzero common zero of K and K' would make both the integral in (4.15) and $w^r(1 - w)^{mr}$ vanish; the only candidate is $w = 1$, where the beta integral is nonzero.

The generic DVR of $P=0$ has uniformizer P and residue field $\mathbb{C}(Q,R)$. Every branch below is unramified, with P also an upstairs uniformizer:

chart	residual equation	(e,f)	position
$U=PS$	$C\int_0^S(1-tQ^m)^r\,dt-R=0$	$(1,r+1)$	$B=0$ affine
$U=Qw_q$	$K_{m,r}(w_q)=0$	$(1,1)$	$A=0$ affine
$U=Qw,\ w\neq w_q$	$K_{m,r}(w)=0$	$(1,1)$	boundary.

(4.16)

The first residual polynomial is irreducible over $\mathbb{C}(Q,R)$ because it is the generic equation $f(S)-R$ of degree $r+1$. The distinguished root is

$$w_q=-\frac{q}{1-q}, \tag{4.17}$$

where $q=h(0)$ is the selected cancellation parameter; the cancellation identity gives $K_{m,r}(w_q)=0$. It is precisely the affine divisor $A=0$. There is no further affine divisorial branch: in the UFD $\mathbb{C}[x,y,z]$ the equation $P=AB$ has exactly the two prime factors A and B. Here $A=1+xy^m$ is primitive and linear in x, while $B=A^{r+1}z+y^{m+1}h(A)$ is primitive and linear in z and is not divisible by A because $h(0)=q\neq 0$. Therefore the other $mr-1$ simple roots are boundary primes, and

$$(r+1)+1+(mr-1)=N. \tag{4.18}$$

Completed generic local rings. The preceding calculations can be summarized without reference to the root charts. If $k_Z=k(Z)$, then $\widehat{\mathcal{O}}_{Y,\eta_Z}=k_Z[[\pi_Z]]$, and the completed normalization at every prime $E\mid Z$ is $k(E)[[u_E]]$. The four cases are:

Z	$\pi_Z\mapsto \text{unit}\cdot u_E^e$	$k(E)$	position
Δ_H	u_E^2	k_Z	boundary critical prime
$C=0$	C	$k_Z[u]/(h_2u^2-Bu+cA)$, $k_Z,\ k_Z$	$W=0$ affine, $W=1$ affine, $W=\rho_j$ boundary
$\Delta_{m,r}$	u_E^{r+1}	k_Z	boundary critical prime
$P=0$	P	$k_Z[S]/(C\int_0^S(1-tQ^m)^r\,dt-R)$, $k_Z,\ k_Z$	$B=0$ affine, $A=0$ affine, $K_{m,r}$-roots boundary.

(4.19)

In the two discriminant rows, every irreducible factor of the residual simple-root cofactor supplies an additional affine completed DVR with $e=1$ and residue field the corresponding factor field. This accounts for every generic DVR extension, including those which do not meet the boundary.

Support and absence of residual components. On $D(C\delta_H)$ in the weighted case and $D(P\delta_{m,r})$ in the cancellation case, every inverse root is simple and every reconstruction coordinate is regular. Thus every boundary image lies in the two listed divisors. Equations (4.4), (4.8), (4.12), and (4.18) saturate (4.1), so no additional normalization prime can lie over either one. Clearing C or P cannot hide a residual vertical component: its generic DVR contribution would be an additional positive summand in a sum already equal to N.

The distinguished source is affine and open in the normal affine finite normalization. Its complement is pure of codimension one. Indeed, at the generic point of a hypothetical higher-codimension irreducible component, choose an affine neighborhood avoiding the divisorial components. Its intersection with the affine source is affine because the ambient scheme is separated. Normal Hartogs extension gives the same coordinate ring on the neighborhood and on that intersection, forcing them to be equal, a contradiction. A height-one prime in the finite integral cover contracts to a nonzero height-one target prime: after excluding contraction to zero, going down over the normal target gives equality of heights. Hence no higher-codimension boundary component or image is omitted.

Over a nonclosed coefficient field, factoring the residual quadratic in (4.5) and $K_{m,r}$ in (4.16) groups geometric branches into arithmetic primes. The factor degrees are exactly their residue degrees, and (4.1) again rules out a new prime after descent. This proves the complete boundary list and all asserted labels. Since $mr = 1$ only for $(m, r) = (1, 1)$, both second target vertices occur throughout the range $N \geq 4$.

The full intersection diagram. Put $p(w) = w(1 - w)^m$ and define the second degree-mr polynomial

$$L_{m,r}(w) = w^{-(r+1)} \int_0^w \{p(w) - p(v)\}^r \, dv. \tag{4.20}$$

Let $S = \{P = Q = 0\}$, with R free. The normalized contraction calculation restricts the critical prime to

$$Q^m L_{m,r}(w) \quad \text{in} \quad \mathbb{C}(R)[w]/(K_{m,r})[[Q]]. \tag{4.21}$$

Consequently, whenever

$$\mathrm{Res}_w(K_{m,r}, L_{m,r}) \neq 0, \tag{4.22}$$

the coefficient of Q^m is a unit in the finite reduced branch algebra. After strict henselization every geometric K-branch therefore intersects the critical prime in

$$\mathbb{C}(R)[[Q]]/(Q^m). \tag{4.23}$$

One of these mr branches is the affine divisor $A = 0$ and the remaining $mr - 1$ are boundary primes. Thus the critical prime has total contact $m^2 r$ with the complete K-cluster and contact $m(mr - 1)$ with the boundary part. Two distinct K-branches force $Q = 0$ and meet in the reduced stratum S; adding the critical prime or any further K-branch does not thicken that intersection. These are all intersections among boundary primes.

Equation (4.22) is an exact, decidable certificate rather than a genericity shortcut. Under $w = -q/(1 - q)$, $K_{m,r}$ is a nonzero scalar multiple of the parameter polynomial $M_{m,r}$ after clearing $(1 - q)^{mr}$. Moreover K and L have the same constant term $1/(r + 1)$, whereas their linear terms are respectively

$$-\frac{mr}{r + 2}, \qquad -\frac{mr(r + 3)}{(r + 1)(r + 2)}.$$

They are therefore not associates. Every proved irreducibility case for $M_{m,r}$ consequently implies (4.22). This proves the diagram for all $mr \leq 30$, the entire column $m = 1$, and the other uniform arithmetic criteria; direct resultants are additionally checked for $1 \leq m, r \leq 5$. In particular every rung of the quadratic stationary-point ladder is covered.

There are also four uniform columns which do not use irreducibility. Put $M_k = K_{m,k}$, including $M_0 = 1$. Scaling $v = wt$ and expanding gives

$$L_{m,r} = \sum_{k=0}^{r} (-1)^k \binom{r}{k} (1 - w)^{m(r-k)} M_k. \tag{4.24}$$

For $r = 1$, this says $L_{m,1} + K_{m,1} = (1 - w)^m$; a common root would be $w = 1$, whereas $K_{m,1}(1) = 1/((m + 1)(m + 2))$. In fact

$$\mathrm{Res}(K_{m,1}, L_{m,1}) = ((m + 1)(m + 2))^{-m}. \tag{4.25}$$

For $r = 2$, set $y = 1 - w$ and $z = y^m$. The endpoint substitution $u = 1 - wt$ gives

$$M_k = \frac{1}{(1-y)^{k+1}} \left\{ \frac{k!}{\prod_{j=1}^{k+1}(mk+j)} - yz^k \sum_{j=0}^{k} \frac{(-1)^j \binom{k}{j} y^j}{mk+j+1} \right\}. \tag{4.26}$$

At a common root, (4.24) and $M_2 = 0$ force $2M_1 = z$, or

$$z\{m(m+1)y^2 - 2m(m+2)y + (m+1)(m+2)\} = 2. \tag{4.27}$$

Substitution in $M_2 = 0$ and clearing nonzero denominators factors as

$$(m+1)^2(y-1)^3\{m^2y - (m+2)^2\} = 0. \tag{4.28}$$

The first root is $w = 0$, where $K_{m,2} = 1/3$. At the second, $y = (m+2)^2/m^2 > 1$, while the braced factor in (4.27) equals $2(m+2)(5m^2+10m+4)/m^3 > 2$; since $z = y^m > 1$, (4.27) is impossible. Hence (4.22) holds for every m when $r = 2$ as well.

For $r = 3$, first note the complementary incomplete-beta identity

$$K_{m,r}(1-y) = \beta_r \sum_{j=0}^{mr} \binom{j+r}{r} y^j, \qquad \beta_r = \frac{r!}{\prod_{j=1}^{r+1}(mr+j)}. \tag{4.29}$$

The consecutive coefficient ratios are $j/(j+r)$, so the Enestrom–Kakeya theorem puts every zero in

$$|y| \le \frac{mr}{mr+r} = \frac{m}{m+1}. \tag{4.30}$$

At a common $K_{m,3}, L_{m,3}$ root, $M_3 = 0$, and (4.24), after division by the nonzero z, becomes $z^2 - 3zM_1 + 3M_2 = 0$. Put $D = 1 - y$ and let T_k denote the sum in (4.26). Clearing the endpoint denominators gives

$$az^2 + bz + c = 0, \qquad dz^3 + e = 0, \tag{4.31}$$

where

$$\begin{aligned} a &= D^3 + 3yDT_1 - 3yT_2, \quad b = -3\beta_1 D, \qquad c = 3\beta_2, \\ d &= -yT_3, \qquad\qquad\qquad\quad\; e \qquad\qquad\; = \beta_3. \end{aligned} \tag{4.32}$$

Their resultant is

$$a^3e^2 + 3abcde - b^3de + c^3d^2. \tag{4.33}$$

It factors as $(y-1)^3 H_m(y)$ up to a nonzero element of $\mathbb{Q}(m)$, where H_m has degree six. Put $\rho = m/(m+1)$ and

$$Q_m(u) = u^6 H_m(\rho/u) = q_0u^6 + q_1u^5 + \cdots + q_6. \tag{4.34}$$

Let A_m, B_m be the lower triangular Toeplitz matrices with entries $(A_m)_{ij} = q_{i-j}$ and $(B_m)_{ij} = q_{6-i+j}$ for $0 \le j \le i \le 5$. After clearing a common denominator in Q_m, the six leading principal minors of the Schur–Cohn matrix

$$A_m A_m^\mathsf{T} - B_m B_m^\mathsf{T} \tag{4.35}$$

are polynomials in m of degrees $29, 56, 81, 104, 125, 144$, respectively, and every coefficient of every minor is strictly positive. The matrix is therefore positive definite for $m \ge 1$. Schur–Cohn gives

$$H_m(y) = 0 \quad \Longrightarrow \quad |y| > \frac{m}{m+1} \tag{4.36}$$

which is disjoint from (4.30). Thus (4.22) also holds for every m when $r = 3$. The exact coefficient-positivity certificate is reproduced by `scripts/verify_contact_resultant_endpoint_reduction.py`.

For $r = 4$, a common root similarly gives $M_4 = 0$ and

$$z^3 - 4z^2 M_1 + 6zM_2 - 4M_3 = 0. \tag{4.37}$$

After clearing endpoint denominators, these are a cubic $E_m(y, z)$ and a quartic $F_m(y, z)$. Their last two subresultants are linear in z and

$$\mathrm{Res}_z(E_m, F_m) = (y - 1)^5 H_m(y), \tag{4.38}$$

up to a nonzero element of $\mathbb{Q}(m)$, where H_m has degree eleven. For $1 \leq m \leq 21$, imposing $z = y^m$ gives the exact gcd $(y - 1)^5$, so only the already excluded endpoint remains.

For $m \geq 22$, the Schur–Cohn form of $u^{11} H_m((m/(m + 1))/u)$ has inertia $(9, 2)$. Indeed, its eleven leading minors have degrees

$$65, 128, 189, 248, 305, 360, 413, 464, 513, 560, 605$$

and, after $m = n + 22$, coefficient signs $+, +, -, +, +, +, +, +, +, +, +$. Thus exactly one conjugate pair of H_m-roots lies in (4.30). With $t = 1/m$ and $x = m(y - 1)$, a rational Rouché certificate localizes its upper member in

$$\left| x + \frac{523}{200} - \frac{41}{8} i \right| < \frac{1}{2}. \tag{4.39}$$

The certificate covers $0 \leq t \leq 1/22$ by 228 rational intervals; exact distance, coefficient, and parameter-derivative bounds prove every strict Rouché inequality.

The disk (4.39) lies inside (4.30) and satisfies $\pi < m \arg(y) < 2\pi$, so y^m lies in the lower half-plane. The linear subresultant recovers $z = -B_m/A_m$. On the rectangular box containing (4.39), every tensor-product Bernstein coefficient of the polynomial equivalent to $-|A_m|^2 \operatorname{Im}(z)$ is strictly negative; hence $A_m \neq 0$ and $\operatorname{Im}(z) > 0$. This contradicts $z = y^m$; the lower disk gives the conjugate contradiction. Therefore (4.22) holds for every m also when $r = 4$. The exact inertia, Rouché, angle, and Bernstein certificates are reproduced by `scripts/verify_contact_resultant_r4.py`. □

5 Stable functoriality

The following proposition records the normalization statement underlying stabilization. It is independent of the two explicit families.

Proposition 5.1 (normalization and boundary under stabilization)**.** *Let Y be a normal integral affine variety over $\mathbb{C}$, let $L/k(Y)$ be a finite extension, and put $X = \mathrm{Norm}_Y(L)$. Let $U \subset X$ be a dense open and $\partial = (X \setminus U)_{\mathrm{red}}$. For $s \geq 0$, write*

$$Y_s = Y \times \mathbb{A}^s, \qquad U_s = U \times \mathbb{A}^s, \qquad L_s = L(t_1, \ldots, t_s).$$

Then there is a canonical isomorphism over Y_s

$$\mathrm{Norm}_{Y \times \mathbb{A}^s}\big(L(t_1, \ldots, t_s)\big) \simeq X \times \mathbb{A}^s. \tag{5.1}$$

In particular, taking $L = k(U)$ and $X = \overline{X}_F$ gives

$$\mathrm{Norm}_{Y \times \mathbb{A}^s}\big(k(U)(t_1, \ldots, t_s)\big) \simeq \overline{X}_F \times \mathbb{A}^s.$$

Under (5.1) the following statements hold.

1. *The reduced boundary is*
$$(X \times \mathbb{A}^s \setminus U_s)_{\mathrm{red}} = \partial \times \mathbb{A}^s, \tag{5.2}$$
and its prime divisors are exactly $E \times \mathbb{A}^s$*, with* E *a prime divisor of* ∂*.*

2. *If a boundary prime* E *lies over a target prime divisor* Z*, the corresponding stabilized primes are* $E_s = E \times \mathbb{A}^s$ *and* $Z_s = Z \times \mathbb{A}^s$*. Their divisorial valuations are the Gauss extensions*
$$v_{E_s}|_L = v_E, \qquad v_{E_s}(t_i) = 0,$$
and
$$e(E_s/Z_s) = e(E/Z), \qquad f(E_s/Z_s) = f(E/Z). \tag{5.3}$$

3. *Every reduced or scheme-theoretic multiple intersection commutes with stabilization. In particular, for target divisors* $Z_1, \ldots, Z_q$ *and boundary divisors* $E_1, \ldots, E_p$*,*
$$\bigcap_i (Z_i \times \mathbb{A}^s) = \left(\bigcap_i Z_i\right) \times \mathbb{A}^s, \qquad \bigcap_j (E_j \times \mathbb{A}^s) = \left(\bigcap_j E_j\right) \times \mathbb{A}^s \tag{5.4}$$
as schemes.

4. *If* R *is the affine coordinate ring of one of these intersections, then the stabilized ring is* $R[t_1, \ldots, t_s]$ *and*
$$\mathrm{Nil}(R[t_1, \ldots, t_s]) = \mathrm{Nil}(R)[t_1, \ldots, t_s]. \tag{5.5}$$
Consequently reducedness and the exact nilpotency index of the nilradical are preserved.

Proof. Write $Y = \operatorname{Spec} A$. If B is the integral closure of A in L, then $X = \operatorname{Spec} B$. The ring $B[t_1, \ldots, t_s]$ is finite and integral over $A[t_1, \ldots, t_s]$, has fraction field L_s, and is normal because a polynomial ring over a normal domain is normal. If an element of L_s is integral over $A[t_1, \ldots, t_s]$, it is also integral over $B[t_1, \ldots, t_s]$; normality then puts it in $B[t_1, \ldots, t_s]$. Thus this polynomial ring is the full integral closure, proving (5.1).

Let $I \subset B$ be the radical ideal defining ∂. The ideal of $\partial \times \mathbb{A}^s$ is $IB[t_1, \ldots, t_s]$, and it is radical because its quotient $(B/I)[t_1, \ldots, t_s]$ is reduced. If $I = \bigcap_\alpha \mathfrak{p}_\alpha$ is its minimal-prime decomposition, then
$$IB[t_1, \ldots, t_s] = \bigcap_\alpha \mathfrak{p}_\alpha B[t_1, \ldots, t_s].$$
Each extended ideal is prime, since $(B/\mathfrak{p}_\alpha)[t_1, \ldots, t_s]$ is a domain, and its height equals the height of $\mathfrak{p}_\alpha$. This proves (5.2) and the asserted bijection of reduced boundary primes.

Let $\mathfrak{p}_E \subset B$ and $\mathfrak{p}_Z \subset A$ be the height-one primes of E and Z. Their extensions are the height-one primes of E_s and Z_s. Since normal height-one local rings are DVRs, their uniformizers may be chosen to be the same as before polynomial extension. At the generic points the residue fields are
$$\kappa(E_s) = \kappa(E)(t_1, \ldots, t_s), \qquad \kappa(Z_s) = \kappa(Z)(t_1, \ldots, t_s).$$
The extended valuation is therefore the Gauss valuation, so its value group on $k(Y)^\times$ is unchanged and every t_i has value zero. The defining identity $v_E|_{k(Y)} = e(E/Z)v_Z$ gives the same ramification index after extension, while
$$[\kappa(E_s) : \kappa(Z_s)] = [\kappa(E) : \kappa(Z)]$$
gives the same residue degree. This proves (5.3).

Scheme intersection is defined by the sum of ideal sheaves. On an affine chart, if the relevant ideals are $J_1, \dots, J_q$, then after stabilization their sum is $(J_1 + \cdots + J_q)[t_1, \dots, t_s]$, and the quotient is

$$\big(A/(J_1 + \cdots + J_q)\big)[t_1, \dots, t_s].$$

This proves (5.4), both downstairs and upstairs; taking radicals also proves the reduced version.

Finally, put $N = \mathrm{Nil}(R)$. The ring $(R/N)[t_1, \dots, t_s]$ is reduced, so every nilpotent polynomial has all coefficients in N. Conversely, the finitely many nilpotent coefficients of a polynomial generate a nilpotent ideal, so every element of $N[t_1, \dots, t_s]$ is nilpotent. This proves (5.5). Moreover

$$\big(N[t_1, \dots, t_s]\big)^a = N^a[t_1, \dots, t_s]$$

for every a. Hence the least $a \geq 1$ for which the nilradical has zero a-th power is unchanged, including the convention $a = 1$ in the reduced case. $\square$

Lemma 5.2 (stable boundary signature)**.** *For the maps in Lemma 4.2, the pair (e_Δ, μ) of (2.1) is invariant under polynomial left–right equivalence and after adjoining any number of identity variables.*

Proof. A left–right equivalence identifies the corresponding finite function-field extensions over isomorphic targets. Uniqueness and functoriality of normalization identify the finite covers, carry the distinguished affine open to the distinguished affine open, and preserve boundary valuations, their e, f labels, and scheme-theoretic target intersections. The intrinsic ramification marking from Lemma 4.2 prevents the two target vertices from being exchanged.

For stabilization, Proposition 5.1 identifies the normalization and the distinguished open pair, preserves all reduced boundary primes and their valuation labels, and base-changes the marked target intersection scheme. Its final assertion preserves reducedness and the exact nilpotency index. Thus both entries of (e_Δ, μ) are unchanged. $\square$

6 Faithfulness of the cancellation parameter

For fixed (m, r), the inverse polynomial (6.2) below is independent of the cancellation root q. The root instead selects the unique nonzero projective branch over $P = 0$ which belongs to the affine reconstruction open. This section proves that the finite normalization together with that open remembers the selection, even after stabilization. The proof uses the reconstruction pole only through its intrinsic boundary residue.

For a normalized cancellation jet write

$$h_q(A) = q + h_1 A + \cdots + h_r A^r$$

and denote the resulting map by F_q. Recall

$$A = 1 + xy^m, \qquad B = A^{r+1} z + y^{m+1} h_q(A), \qquad P = AB, qquad Q = y + xB. \tag{6.1}$$

The resolvent is

$$\Psi_{P,Q,R}(T) = C \int_0^T \{1 - t(Q - Pt)^m\}^r \, dt - R. \tag{6.2}$$

We first record the part of the target-boundary calculation which remains valid in the presence of arbitrary stable variables.

Lemma 6.1 (stable boundary-plane scaling)**.** *Let $\boldsymbol{\tau}$ be any finite tuple of variables. Suppose a polynomial automorphism of*

$$\mathbb{A}^3_{P,Q,R} \times \mathbb{A}^{|\boldsymbol{\tau}|}$$

preserves the two labelled cancellation boundary divisors $P = 0$ and $\Delta_{m,r} = 0$. Then for some $a, u \in k^*$ its ring pullback satisfies

$$\beta^* P = aP, \qquad \beta^* Q \equiv uQ \pmod{P}, \qquad \beta^* R \equiv u^{-m} R \pmod{P}. \tag{6.3}$$

The conclusion permits arbitrary mixing of the stable variables.

Proof. Preservation of the prime divisor $P = 0$ gives $\beta^* P = aP$, since the only units of the polynomial ring are constants. On the boundary plane the scheme-theoretic intersection with the discriminant is

$$Q^M G = 0, \qquad M = mr(m+1), \qquad G = (r+1)RQ^m - C. \tag{6.4}$$

After adjoining $\boldsymbol{\tau}$, the two reduced components have coordinate rings

$$k[R, \boldsymbol{\tau}] \quad \text{and} \quad k[Q, Q^{-1}, \boldsymbol{\tau}]. \tag{6.5}$$

Their unit groups modulo k^* have ranks zero and one, respectively, so an automorphism cannot exchange them. The induced automorphism $\bar{\beta}$ of $k[Q, R, \boldsymbol{\tau}]$ therefore preserves the prime ideals (Q) and (G). Hence

$$\bar{\beta}(Q) = uQ, \qquad \bar{\beta}(G) = vG \tag{6.6}$$

for $u, v \in k^*$. Substitution of the first equality into the second gives

$$(r+1)u^m \bar{\beta}(R) Q^m - C = v\{(r+1)RQ^m - C\}.$$

Reduction modulo Q gives $v = 1$, and cancellation of Q^m then gives $\bar{\beta}(R) = u^{-m} R$. This proves (6.3). $\square$

Theorem 6.2 (stable cancellation-parameter faithfulness)**.** *Let k be an algebraically closed field of characteristic zero, fix $m, r \geq 1$, and let q, q' be roots of $M_{m,r}$. If the normalized cancellation maps F_q and $F_{q'}$ are stably polynomially left–right equivalent, then*

$$q = q' \qquad \text{and} \qquad h_q = h_{q'}. \tag{6.7}$$

Equivalently, the common finite normalization of (6.2)*, together with its distinguished affine reconstruction open and its map to the target, faithfully remembers the selected cancellation root.*

Proof. The case $mr = 1$ has only one parameter root, so assume $mr > 1$. Then $P = 0$ is the intrinsically labelled unramified target-boundary vertex. After adjoining the same tuple of identity variables to both maps, write the ring identity supplied by a left–right equivalence as

$$\alpha^* F^*_{q'} = F^*_q \beta^*. \tag{6.8}$$

Lemma 6.1 gives

$$\beta^* P = aP, \qquad \beta^* Q = uQ + P\varphi \tag{6.9}$$

for $a, u \in k^*$ and a polynomial φ, which may involve the stable variables.

In the affine reconstruction open, the inverse image of $P = 0$ has exactly the two prime components $A = 0$ and $B = 0$. Their generic residue degrees over $P = 0$ are respectively 1 and $r + 1$, by the projective boundary chart (4.16). These labels survive stabilization, so (6.8) cannot exchange the two components. Unique factorization and (6.9) therefore give a constant $c \in k^*$ such that

$$\alpha^* A' = cA, \qquad \alpha^* B' = \frac{a}{c} B. \tag{6.10}$$

The scalar c is forced to be one. Indeed, in the stabilized source the fiber $A = 1$ has coordinate ring

$$k[x, y, z, \boldsymbol{\xi}]/(xy^m), \tag{6.11}$$

which is not a domain, whereas for every $t \neq 1$ the fiber $A = t$ has coordinate ring

$$k[y, y^{-1}, z, \boldsymbol{\xi}] \tag{6.12}$$

after solving $x = (t-1)y^{-m}$, and is a domain. The pullback of the unique nondomain fiber $A' = 1$ under (6.10) is $A = c^{-1}$; hence $c = 1$.

Put

$$X = \alpha^* x', \qquad Y = \alpha^* y'.$$

Since $\alpha^* A' = A$, equation $A' - 1 = x'(y')^m$ becomes

$$XY^m = xy^m. \tag{6.13}$$

The stabilized source ring is a UFD, and X, Y are nonassociate primes. If $m > 1$, comparison of prime multiplicities in (6.13) gives

$$X = \lambda^{-m} x, qquad Y = \lambda y \tag{6.14}$$

for some $\lambda \in k^*$. If $m = 1$, unique factorization also permits the formal interchange $X \sim y$, $Y \sim x$. It cannot occur: reducing the Q-identity in (6.8) modulo B gives $Y \equiv uy \pmod{B}$, while

$$(B) \cap k[x, y, \boldsymbol{\xi}] = 0 \tag{6.15}$$

because $B = A^{r+1}z + y^{m+1}h_q(A)$ has positive z-degree. Thus the interchange would make a nonzero linear combination of x and y belong to (B). Consequently the scaling in (6.14) holds for every m. The same reduction modulo B now gives $\lambda = u$.

Using (6.10) with $c = 1$, the full Q-identity is

$$uy + au^{-m}xB = u(y + xB) + AB\, F_q^*(\varphi). \tag{6.16}$$

Cancel B. If $au^{-m} - u$ were nonzero, (6.16) would imply $A \mid x$, which is false. Therefore

$$a = u^{m+1}. \tag{6.17}$$

At the affine prime $A = 0$, the function y is a unit and

$$\left.\frac{B}{y^{m+1}}\right|_{A=0} = q. \tag{6.18}$$

Equations (6.10) and (6.14)–(6.17) carry the analogous residue for $F_{q'}$ to

$$\left.\frac{u^{m+1}B}{(uy)^{m+1}}\right|_{A=0} = q.$$

Thus $q' = q$. The finite cancellation recursion gives a unique normalized degree-r jet over each parameter root, so $h_{q'} = h_q$ as well. $\square$

Remark 6.3 (reconstruction-pole interpretation)**.** *On the inverse chart $D = A^{-1}$, $B = PD$, and $y = Q - sP$, so (6.18) is*

$$q = \mathrm{res}_{A=0} \frac{PD}{(Q - sP)^{m+1}}. \tag{6.19}$$

Equivalently, q is the first coefficient after the canonical leading pole in

$$PD^{r+2} - z = (Q - sP)^{m+1} D^{r+1}\{q + h_1 D^{-1} + \cdots + h_r D^{-r}\}. \tag{6.20}$$

The proof above is the coordinate-free normalization needed before this Laurent coefficient can be used: the marked open recovers A from the two affine factors of P, fixes its scalar by the unique nondomain fiber, and recovers y up to the same scaling which acts on B with weight $m + 1$.

Corollary 6.4 (distinct roots give distinct stable classes)**.** *For fixed (m, r), the mr geometric roots of the squarefree polynomial $M_{m,r}$ give mr pairwise stably polynomially left–right inequivalent cancellation maps. Cancellation maps belonging to different pairs (m, r) of the same inverse degree remain separated by their boundary signatures. Consequently, if $n = N - 1$, the cancellation branches together with one weighted class give the degreewise lower bound*

$$1 + \sum_{\substack{r|n \\ r<n}} (n - r) = 1 + n\tau(n) - \sigma(n).$$

7 Reduced and thick boundary contact

We isolate the cancellation calculation because it supplies the numerical invariant used in the degreewise theorem. The first proof is global and eliminative; the second is local and computes the contact order without a resultant.

Proposition 7.1 (all-parameter cancellation trace)**.** *For every $m, r \geq 1$, let $\delta_{m,r}$ be the reduced prime discriminant equation fixed in Section 4. Up to multiplication by a nonzero scalar,*

$$\delta_{m,r}(0, Q, R) = Q^{mr(m+1)}\big((r + 1)RQ^m - C\big). \tag{6.1}$$

Consequently

$$\frac{\mathbb{C}[Q, R]}{(\delta_{m,r}(0, Q, R))} \simeq \frac{\mathbb{C}[Q, R]}{(Q^{mr(m+1)})} \times \mathbb{C}[Q, Q^{-1}], \tag{6.2}$$

and the nilradical has exact nilpotency index $mr(m + 1)$.

First proof: monic resultant and subresultant. On the normalized critical divisor put $Y = Q - PT$. Then

$$T = Y^{-m}, \quad P = (Q - Y)Y^m. \tag{6.3}$$

After substituting $t = uY^{-m}$ in the critical-value equation and clearing the unit $Y^{-m(r+1)}$, define

$$F(Y) = P - (Q - Y)Y^m, \tag{6.4}$$

$$G(Y) = CI_{m,r}(Y, Q) - RY^{m(r+1)}, \tag{6.5}$$

where

$$I_{m,r}(Y, Q) = \int_0^1 [Y^m - u\{Q + (Y - Q)u\}^m]^r \, du. \tag{6.6}$$

Equivalently, the integral-free coefficient formula is

$$I_{m,r} = \sum_{j=0}^{r}\sum_{k=0}^{mj} \frac{(-1)^j}{j+k+1}\binom{r}{j}\binom{mj}{k} Y^{m(r-j)} Q^{mj-k}(Y-Q)^k.$$

Thus (6.6) is a polynomial with rational coefficients. The monic polynomial F is primitive over $\mathbb{Z}[C,P,Q,R]$. Multiplication of G by $(mr+r+1)!$ clears every integration denominator; taking its primitive part gives an integral polynomial with the same zero scheme over $\mathbb{C}$. This works uniformly for every symbolic pair of positive integers (m,r). It is the precise uniform integral meaning here: m and r occur as exponents, so there is no literal polynomial ring in indeterminates m,r containing all members at once.

The polynomial F is monic in Y. Its incidence with G is the integral critical-value incidence of Section 4 and maps birationally to the reduced discriminant. At its generic point, F and G have exactly one common simple root: the critical points are simple and their critical values are distinct. Thus the zeroth subresultant specializes to zero and the first subresultant does not, certifying a gcd of degree one. More concretely, after adjoining the distinct roots of F, the resultant is the product of the G-values; generically exactly one factor vanishes, and it is transverse because G is linear in R with nonzero coefficient at that root. It follows that the primitive resultant

$$D_{m,r}(P,Q,R) = \mathrm{Res}_Y(F,G) \tag{6.7}$$

is a nonzero scalar multiple of the reduced prime generator $\delta_{m,r}$, not a power of it.

Now specialize $P=0$. Since $F(Y) = Y^m(Y-Q)$ is monic,

$$D_{m,r}(0,Q,R) = G(0)^m G(Q). \tag{6.8}$$

The two values of (6.6) are uniform beta integrals:

$$I_{m,r}(0,Q) = (-1)^r Q^{mr}\frac{r!(mr)!}{(mr+r+1)!}, \tag{6.9}$$

$$I_{m,r}(Q,Q) = \frac{Q^{mr}}{r+1}. \tag{6.10}$$

Hence $G(0)$ is a nonzero scalar times CQ^{mr}, while

$$G(Q) = Q^{mr}\left(\frac{C}{r+1} - RQ^m\right). \tag{6.11}$$

Substitution in (6.8) gives (6.1). This derivation takes the resultant only after F,G have been made polynomial and primitive; no component can be lost through denominator clearing.

The two factors in (6.1) are comaximal because the second is congruent to $-C$ modulo Q. The Chinese remainder theorem gives (6.2). In the first factor the nilradical is generated by Q and has exact index $mr(m+1)$; the second factor is reduced. □

Second proof: completed local contact. This proof computes the exponent in (6.1) without using a resultant. Let η be the generic point of the affine-line component $P=Q=0$ of the intersection, and put $K=\mathbb{C}(R)$. The completed target local ring transverse to that line is

$$\widehat{\mathcal{O}}_{\mathbb{A}^3,\eta} = K[[P,Q]]. \tag{6.12}$$

Use the completed regular parameter plane $A = K[[Y,Q]]$ for the polynomial critical-value incidence; its slice over the generic value R is $G = 0$. The pullback of the plane $P = 0$ is the Cartier divisor

$$P = (Q - Y)Y^m, \qquad \operatorname{div}(P) = m[Y = 0] + [Y = Q]. \tag{6.13}$$

Neither Y nor $Q - Y$ divides G, by (6.9)–(6.10). Additivity of intersection multiplicity for the divisor (6.13) therefore gives

$$\operatorname{length}_K A/(P, G) = m \operatorname{ord}_Q G(0, Q) + \operatorname{ord}_Q G(Q, Q). \tag{6.14}$$

Equations (6.9) and (6.11), now used only as completed-local restrictions, give

$$\operatorname{ord}_Q G(0, Q) = mr, \quad \operatorname{ord}_Q G(Q, Q) = mr, \tag{6.15}$$

because $C/(r + 1) - RQ^m$ is a unit in $K[[Q]]$. Consequently

$$\operatorname{length}_K A/(P, G) = m(mr) + mr = mr(m + 1). \tag{6.16}$$

The critical-value normalization is birational with residue degree one. The local projection formula therefore identifies (6.16) with the order of $\delta_{m,r}(0, Q, R)$ in $K[[Q]]$. Thus the completed intersection ring at η is

$$K[[Q]]/(Q^{mr(m+1)}), \tag{6.17}$$

which directly gives contact order and nilpotency index $mr(m + 1)$. The two summands in (6.14) also explain the exponent geometrically: m^2r comes from the multiplicity-m branch $Y = 0$, and mr from the transverse branch $Y = Q$. □

Lemma 7.2 (contact calculation). *For the split weighted map, $Z_\Delta \cap Z_0$ is reduced and $\mu = 1$. For a cancellation map of type (m, r) with $N \geq 4$, it is nonreduced and*

$$e_\Delta = r + 1, \qquad \mu = mr(m + 1) = m(N - 1) > 1. \tag{6.18}$$

Proof. For the weighted seed, Lemma 4.1 gives

$$C = 0, \qquad \delta_N|_{C=0} = \mu_N(B^2 - 8A) = 0,$$

with $\mu_N \neq 0$. Its coordinate ring is $\mathbb{C}[B]$, hence is reduced.

For cancellation, $e_\Delta = r + 1$ was computed in Lemma 4.2, and Proposition 7.1 gives the scheme intersection and its exact nilpotency index. In the range $N \geq 4$, $P = 0$ is a canonical boundary vertex by Lemma 4.2, and the index exceeds one. □

8 The scheme-theoretic decorated normalization

For a weighted primitive H, let D_H be the reduced discriminant of $H(W) - sW + t$ and let

$$\nu_H : \mathbb{A}^1_r \longrightarrow D_H, \qquad r \longmapsto (H'(r), rH'(r) - H(r)) \tag{6.1}$$

be its affine normalization. Write $R_H = k[H'(r), rH'(r) - H(r)] \subset k[r]$.

Proposition 8.1 (full Fitting divisor). *There is an equality of ideal sheaves on the normalization*

$$\operatorname{Fitt}_0 \Omega_{\widetilde{D}_H/D_H} = (H''(r)). \tag{6.2}$$

This ideal, including every root multiplicity, is preserved by polynomial left–right equivalence and by adjoining identity variables.

Proof. The images in $k[r]\,dr$ of the two generators of R_H are

$$dH'(r) = H''(r)dr, \qquad d(rH'(r) - H(r)) = rH''(r)dr.$$

Thus $\Omega_{k[r]/R_H} \simeq k[r]/(H'')\,dr$, proving (6.2). The stable boundary theorem identifies the finite normalization morphisms on the intrinsic discriminant vertex away from the second vertex, so it identifies their relative differential modules and Fitting ideals. Under stabilization one has

$$\Omega_{(\widetilde{D}_H \times \mathbb{A}^q)/(D_H \times \mathbb{A}^q)} \simeq \mathrm{pr}_1^* \Omega_{\widetilde{D}_H/D_H}.$$

Zeroth Fitting ideals commute with this flat base change. Equality of the ideal sheaves preserves their valuations at every height-one point, hence preserves the multiplicities in (6.2), not only the reduced support. □

If the support of (6.2) contains two points, the units of $k[r, \xi, \xi^{-1}, z_1, \ldots, z_q]$ and subtraction of the two corresponding principal-prime equations force the induced coordinate change to have the form $r \mapsto ar + b$, $a \neq 0$. Proposition 8.1 then says that the full effective Hessian divisor is carried affinely, with its coefficients.

The self-intersection of (6.1) is also intrinsic. In two normalization coordinates put

$$D_s(r, u) = \frac{H'(r) - H'(u)}{r - u},$$

$$D_t(r, u) = \frac{rH'(r) - H(r) - uH'(u) + H(u)}{r - u}.$$

The genuine off-diagonal closure is the scheme

$$\Gamma_H^{\mathrm{off}} = V\big((D_s, D_t) : (r - u)^\infty\big). \tag{6.3}$$

The saturation in (6.3) is necessary: on the diagonal the divided equations specialize to $H''(r)$ and $rH''(r)$ and otherwise retain isolated cusp points. The involution $(r, u) \leftrightarrow (u, r)$ is free on the ordinary off-diagonal locus; its characteristic-zero finite quotient exists in general and retains fixed diagonal limits in degenerations. On the ordinary locus that quotient records the two normalization branches paired at each node. An isomorphism of normalization morphisms preserves the fiber product, its diagonal, and the saturated complement, while stabilization takes their product with affine space. Hence this node-pairing scheme is stable. In the exact extractor, ordinary-node status is certified by the Jacobian determinant

$$\det \frac{\partial(D_s, D_t)}{\partial(r, u)}$$

being a unit on (6.3), rather than inferred only from the number of projected branch points. Passing to $\sigma = r + u, \pi = ru$ gives the scheme quotient by the free involution; the implementation verifies that its equations pull back into the saturated ordered ideal.

Finally, let

$$\mathfrak{c}_H = \mathrm{Ann}_{\mathcal{O}_{D_H}}(\nu_{H*}\mathcal{O}_{\widetilde{D}_H}/\mathcal{O}_{D_H}) \tag{6.4}$$

be the conductor. It is functorial for the same reason. On the open locus of ordinary cusps and nodes, if $P_H(r)$ cuts out all normalization branches appearing in (6.3), then the local models $k[[t^2, t^3]] \subset k[[t]]$ and $k[[x, y]]/(xy) \subset k[[x]] \oplus k[[y]]$ give

$$\mathfrak{c}_H k[r] = (H''(r)^2 P_H(r)). \tag{6.5}$$

For arbitrary plane-curve singularities, let $f_H(S,T)=0$ be the primitive implicit equation. Plane-curve adjunction gives the uniform generator

$$\mathfrak{c}_H k[r] = \left(\frac{(\partial f_H/\partial T)(H'(r), rH'(r)-H(r))}{H''(r)}\right). \tag{6.5a}$$

This quotient is polynomial because (6.1) is the normalization. The exact extractor uses (6.5a) without an ordinary-singularity hypothesis and checks (6.5) independently whenever the cusp/node criterion holds. The invariant retains the finite conductor morphism

$$\operatorname{Spec}(k[r]/\mathfrak{c}_H k[r]) \longrightarrow \operatorname{Spec}(R_H/(R_H \cap \mathfrak{c}_H k[r])), \tag{6.6}$$

not only its upstairs support. It records the two branches over a node and the single thick branch over a cusp. Together, (6.2)–(6.6), the point at infinity, and the labeled inverse image of the intersection with the second target boundary form the decorated normalization. The full second-boundary center divisor upstairs is $\gcd(H,H')$; a point label is inferred only when its support is uniquely certified. Every part is therefore preserved under stable polynomial left–right equivalence.

For the split quartics, the boundary label on the normalization is also checked intrinsically. If zero is the unique multiple primitive root and $H(r)=h_2r^2+O(r^3)$, the normalized discriminant boundary chart

$$B=\frac{H'(r)}{C}, \qquad A=\frac{rH'(r)-H(r)}{cC^2}, \qquad r=Ck$$

specializes to

$$B=2h_2k, \qquad A=\frac{h_2}{c}k^2.$$

A finite specialization centered at $r=\rho$ forces $H(\rho)=H'(\rho)=0$, hence $\rho=0$. Thus the second-boundary intersection canonically marks $r=0$; it is not an auxiliary normalization convention.

Proposition 8.2 (nonempty ordinary boundary-clean locus)**.** *For every $N\geq 4$, the normalized admissible degree-N seed space contains a nonempty open on which the discriminant has only ordinary cusps and nodes, zero is the unique multiple primitive root, and the boundary mark $r=0$ belongs to neither the cusp nor the node-branch scheme.*

Proof. The generic discriminant theorem proves that the ordinary nodal-cuspidal open meets the normalized admissible slice in every degree. Boundary-cleanness and avoidance of the finite cusp and node-branch schemes are also open conditions. They are nonempty on the normalized slice, as witnessed by

$$H_N^{\mathrm{split}}(W)=\frac{1}{2}W^2(1-W)(1+W^{N-3}).$$

Indeed,

$$(H_N^{\mathrm{split}})'(1)=-1, \qquad (H_N^{\mathrm{split}})''(1)=-(N+1)\neq -2, \qquad (H_N^{\mathrm{split}})''(0)=1.$$

Put $P_N=2H_N^{\mathrm{split}}/W^2=(1-W)(1+W^{N-3})$. This polynomial is squarefree and has no zero root. At every root α of P_N,

$$\left.\frac{2(H_N^{\mathrm{split}})'(W)}{W}\right|_{W=\alpha}=\alpha P_N'(\alpha)\neq 0.$$

Thus $\gcd(H_N^{\mathrm{split}},(H_N^{\mathrm{split}})')=W$. The nonzero second derivative at zero excludes a cusp there. If an off-diagonal point u had the same discriminant image as zero, then $H'(u)=H(u)=0$, contradicting the gcd calculation. Hence the boundary mark is not a node branch. The normalized seed space is irreducible, so its nonempty ordinary and boundary-clean opens intersect. $\square$

For completeness, this decoration gives an actual generic-dimension theorem, not only a parameter count. In the normalized degree-N seed space, an isomorphism preserving zero and infinity is $r \mapsto ar$. Equality of Fitting divisors then has the form $G''(r) = cH''(ar)$. Since the normalized seeds vanish together with their first derivatives at zero, integration gives

$$G(r) = \frac{c}{a^2} H(ar).$$

The condition $G(1) = 0$ forces $H(a) = 0$, leaving at most $N - 2$ generic choices, and $G'(1) = -1$ determines c. Thus the decorated-normalization moduli map is generically finite and its image has dimension $N - 3$.

Proposition 8.3 (generic unramifiedness and rerooting degree)**.** *For every $N \geq 4$, the decorated-normalization moduli map is generically unramified. After restriction to a nonempty open it is etale of degree $N - 2$ onto the normalization of its image, and its geometric fibers are exactly the rerootings at the nonzero simple roots of the seed.*

Proof. On the squarefree boundary-clean open, retain only the Fitting divisor on $(\P^1; 0, \infty)$. A normalized first-order seed deformation $H + \varepsilon \dot{H}$ which is trivial in the quotient of these divisors by normalization-line scaling satisfies

$$\dot{H}''(r) = \alpha H''(r) + \beta r H'''(r). \tag{6.7}$$

The first term is the scalar ambiguity in a defining section and the second is the infinitesimal action of $r \mapsto (1 + \varepsilon\beta)r$; these are all infinitesimal automorphisms because zero and infinity are marked. Twice integrating and using $\dot{H}(0) = \dot{H}'(0) = 0$ gives

$$\dot{H} = \alpha H + \beta(rH' - 2H).$$

The tangent endpoint equations now give

$$0 = \dot{H}(1) = -\beta, \qquad 0 = \dot{H}'(1) = -\alpha,$$

so $\dot{H} = 0$. The Fitting-divisor quotient, and therefore the full decorated map, is generically unramified. On the locus with trivial divisor stabilizer, the seed space and the Fitting-divisor quotient are smooth of dimension $N - 3$, hence the latter map is etale after shrinking.

For the degree statement, if a is a nonzero simple root of H, put

$$\kappa_a = -\frac{1}{aH'(a)}, \qquad G_a(w) = \kappa_a H(aw).$$

Then G_a is normalized and

$$E_{G_a}(w; s, t) = \kappa_a E_H\left(aw; \frac{s}{\kappa_a a}, \frac{t}{\kappa_a}\right). \tag{6.8}$$

Thus all parts of the decoration, including the node pairing and conductor, are transported. Generically $H = w^2 P$ with P squarefree of degree $N - 2$, and these $N - 2$ rerootings are distinct. Conversely, the integration argument preceding the proposition shows that every point of a decorated fiber has this form. The generic degree is therefore exactly $N - 2$, and the normalization of the decorated image is generically birational to the Fitting-divisor quotient. After one final shrinking they are isomorphic, which proves the asserted etaleness onto the normalized image. $\square$

Lemma 8.4 (low-pole filtration)**.** *Let* $\deg H = N$ *and set* $s = H'(r)$, $t = rH'(r) - H(r)$ *and* $R_H = k[s,t] \subset k[r]$. *For the pole-order filtration at the unique point at infinity of the projective normalization,*

$$L_m(R_H) = \{f \in R_H : \mathrm{ord}_\infty(f) \geq -m\},$$

one has

$$L_{N-2} = k, \qquad L_{N-1} = k \oplus ks, \qquad L_N = k \oplus ks \oplus kt. \tag{6.9}$$

These equalities remain valid after an arbitrary field extension.

Proof. The element r is integral over $k[s]$, while $k(s,t) = k(r)$ and $[k(r) : k(s)] = N-1$. Thus the implicit equation of D_H is monic of degree $N-1$ in t, so each class has a unique representative

$$\sum_{0 \leq j < N-1} p_j(s) t^j.$$

The monomial $s^i t^j$ has pole order $i(N-1) + jN$. If two such orders are equal with $0 \leq j, j' < N-1$, reduction modulo $N-1$ gives $j = j'$ and then $i = i'$. Hence leading poles cannot cancel, and (6.9) follows. The argument is coefficient-field independent. □

Theorem 8.5 (full-cover faithfulness)**.** *Let* k *be a characteristic-zero field. Let* $H, G \in k[W]$ *be normalized degree-*N *seeds:*

$$H(0) = H'(0) = H(1) = 0, \quad H'(1) = -1,$$

and similarly for G. *Suppose zero has multiplicity at least two, with no restriction on collisions or multiplicities among the other roots. If their full marked incidence covers, including the regular-reconstruction opens, are stably left–right equivalent, then* $H = G$. *Consequently,*

$$F_H \sim_{\text{stable LR}} F_G \quad \Longrightarrow \quad H = G. \tag{6.10}$$

Proof. Localize away from the second target boundary and put $K = k(C)$, $s = BC$, and $t = cAC^2$. We first fix the variance convention. If the geometric target map is $\beta : Y_H \to Y_G$, its ring map is

$$\beta^* : K[s_G, t_G, z_1, \ldots, z_q] \longrightarrow K[s_H, t_H, z_1, \ldots, z_q]. \tag{6.11}$$

Thus the repository convention $G = \beta F \alpha$ uses β^{-1} when old target coordinates are expressed in the new presentation, but uses β^* in (6.11). The latter takes an irreducible equation f_G of the discriminant to a unit times f_H.

The curve D_H is geometrically irreducible by the universal-pencil theorem and is singular: H'' is nonconstant, and $d\nu_H$ vanishes at a root of H''. In particular f_H cannot lie in a one-variable coordinate subring, since geometric irreducibility would then make it linear in that coordinate and D_H a smooth coordinate line. Dryło's stable-equivalence theorem for $K[s,t]$ therefore implies that β^* preserves the plane subring $K[s,t]$.[1]

The descended plane automorphism preserves the unique infinity place. Lemma 8.4, applied in both directions, gives on D_H

$$\beta^* s_G = \lambda s_H + b, \qquad \beta^* t_G = \mu t_H + d s_H + e, \tag{6.12}$$

[1] R. Dryło, *On the stable equivalence problem for* $k[x,y]$, Colloq. Math. 124 (2011), 247–253, Theorem 1.

where $\lambda\mu \neq 0$. Every second-boundary center is a multiple primitive root and hence has $s = t = 0$; since zero supplies a center on both sides, this gives $b = e = 0$ without requiring uniqueness. Since $r = dt/ds$ on the normalization, (6.12) induces

$$r_G = \frac{\mu r_H + d}{\lambda};$$

compose with the inverse triangular linear automorphism in (6.12). The resulting plane automorphism fixes D_H pointwise. The fixed-curve theorem of Blanc–Stampfli says that a nonidentity affine-plane automorphism can fix pointwise only a curve equivalent to a line.[2] Since D_H is singular, the composition is the identity. Hence (6.12), with $b = e = 0$, holds in the entire plane ring.

Put $a = \lambda/\mu$ and $p = d/\lambda$. Restricting to the normalization and integrating gives

$$G'(r/a + p) = \lambda H'(r), \qquad G(w) = \mu H(a(w - p)). \tag{6.13}$$

The t identity makes the integration constant zero. The parameters a, p, μ lie in k: the affine map $w \mapsto a(w - p)$ carries the finite root multiset of G to that of H, and the images of the distinct roots $0, 1$ determine a, p up to a finite constant choice; the leading coefficient then determines μ. The pullback orientation is checked without convention by the identity

$$\beta^* E_G(w) = G(w) - \lambda s_H w + \mu t_H + d s_H = \mu E_H(a(w - p); s_H, t_H). \tag{6.14}$$

The universal pencil has monodromy S_N. Its degree-N incidence field corresponds to the self-normalizing subgroup S_{N-1}, so the cover has no nontrivial deck automorphism; purely transcendental stabilization adds none. Therefore the actual cover lift in (6.14) is $W_H = a(W_G - p)$.

Let $m = \mathrm{ord}_0(G) \geq 2$. The Newton polygon of $hW^m - BCW + AC^2$ shows that the zero cluster has affine sheets: both sheets when $m = 2$, and one slope-one sheet when $m \geq 3$. It maps to the H-root $-ap$ with multiplicity m. If $-ap \neq 0$, all sheets of this nonzero multiple-root cluster are polar, since $y = (W - \gamma)/C$ has leading term $(-ap)/C$. Preservation of the reconstruction open forces $p = 0$.

The affine root-one component for G now maps to the simple H-root a. Every simple extra root is polar. If $H'(a) + a \neq 0$, the coordinate $y = (W - \gamma)/C$ has a pole. If that leading pole cancels, then $\gamma \to a$ and $x \to 0$, while the numerator of the reconstruction formula for z tends to $a - 1$. Thus regularity still forces $a = 1$. Now $G = \mu H$, and $G'(1) = H'(1) = -1$ forces $\mu = 1$. □

Remark 8.6. *Theorem 8.5 concerns the full incidence cover with its affine open. Proposition 8.3 is unchanged: the coarser decorated-normalization map forgets that open and remains generically etale of degree $N - 2$. The exact-double-zero boundary-clean locus is its generic special case; higher zero multiplicity and arbitrary extra-root collisions alter the compactified rerooting groupoid but introduce no new full-cover identifications.*

Proposition 8.7 (Hurwitz–LL compactification)**.** *For $P_{H,s}(W) = H(W) - sW$, the curve D_H is the pullback of the universal Lyashko–Looijenga point-in-branch-divisor incidence along $s \mapsto P_{H,s}$. The normalized seed locus has a proper compactification $\overline{\mathcal{S}}_N^{\mathrm{adm}}$ obtained as the normalization of its closure in the stack of marked admissible polynomial covers. The generic degree-$N - 2$ rerooting relation extends to the proper marked/unmarked groupoid. On the collision-separating model it is a finite representable etale quotient-stack morphism. A total collision of m zero-fiber points has m-cycle monodromy on the special transverse slice $\epsilon = \delta^m$, while generic coarse divisorial inertia is a transposition.*

[2] J. Blanc and I. Stampfli, *Automorphisms of the plane preserving a curve*, 2013.

Proof. A critical point r of $P_{H,s}$ satisfies $s = H'(r)$, and its critical value is

$$P_{H,s}(r) = H(r) - rH'(r) = -t.$$

This proves the first assertion and recovers the normalization (6.1). On the exact-double distinct-root open,

$$H(W) = -\frac{W^2(W-1)\prod_{i=1}^{N-3}(W-\rho_i)}{\prod_{i=1}^{N-3}(1-\rho_i)},$$

so the labelled source is the configuration space of the $N-3$ points ρ_i on the thrice-marked rational curve. Its collision-separating compactification is $\overline{M}_{0,N}$; quotienting by S_{N-3} forgets their labels. Equivalently, the seed defines a polynomial Hurwitz cover with total ramification over infinity, a marked ramified point p_0 over zero, and a marked unramified point p_1 over zero. Taking the normalization of its closure in the proper admissible-cover stack gives $\overline{\mathcal{S}}_N^{\mathrm{adm}}$.[3]

Put $n = N-2$. Forgetting which of the n simple zero-fiber points is designated p_1, while retaining all of them as an unordered divisor, is the quotient morphism

$$[\overline{M}_{0,N}/S_{n-1}] \longrightarrow [\overline{M}_{0,N}/S_n]. \tag{6.15}$$

It is finite, representable, and etale of degree $[S_n : S_{n-1}] = n$, and restricts to the rerooting relation (6.8). On the cyclic slice $u^m = \epsilon$, normalization writes $\epsilon = \delta^m$ and cycles the m choices. For $m > 2$ this slice passes through the codimension-$(m-1)$ total-collision locus; the generic coarse discriminant instead has one pair collision and transposition inertia. Finally, the stable-map branch divisor extends in flat families, so the LL incidence extends over the closure.[4] Its finite part has degree $N-1$, after removing the fixed branch contribution $(N-1)[\infty]$. □

Proposition 8.8 (rerooting boundary pullback)**.** *Let π be (6.15). For any boundary divisor whose distinguished component contains exactly k of the n simple zero-fiber marks, write $\Delta_{k,\mathrm{in}}$ and $\Delta_{k,\mathrm{out}}$ according as p_1 belongs to that component or not. Then*

$$\pi^*[\Delta_k] = [\Delta_{k,\mathrm{in}}] + [\Delta_{k,\mathrm{out}}], \tag{6.16}$$

with restriction degrees k and $n-k$. After contraction to the coarse coefficient collision model, the generic discriminant pullback consists of one degree-one component of ramification index two and one unramified component of degree $n-2$.

Proof. The quotient-stack map (6.15) is etale, so both coefficients in (6.16) are one. Above an unordered boundary configuration, selecting one of the k points on the distinguished component gives the first restriction, while selecting one of the remaining $n-k$ gives the second. At a generic coefficient collision, choosing the double value has local square-root normalization and contributes $2 \cdot 1$ to the degree; choosing one of the $n-2$ other values is unramified. Thus the degree sum is $2 + (n-2) = n$. □

Remark 8.9. *The number $N-2$ here is the subgroup index $[S_n : S_{n-1}]$, not the degree of the classical LL morphism. The latter has degree N^{N-2} on the full space of monic centered degree-N polynomials.*[5] *The fixed pencil and the normalized seed locus have smaller dimensions, so an LL degree for their restriction requires a separately specified Hurwitz stratum and intersection problem.*

[3]For the normalization and twisted-cover interpretation of the admissible-cover stack, see D. Abramovich, A. Corti and A. Vistoli, *Twisted bundles and admissible covers*. Collision-allowing intermediate models are constructed in A. Deopurkar, *Compactifications of Hurwitz spaces*.

[4]B. Fantechi and R. Pandharipande, *Stable maps and branch divisors*.

[5]See S. Lando and D. Zvonkine, *Counting ramified coverings and intersection theory on spaces of rational functions I*, and the references there for the classical polynomial LL degree.

Theorem 8.10 (marked-zero-fiber LL degree)**.** *Let $\mathcal{A}_N^\circ$ be the exact-double normalized seed locus on which all critical values are simple. Put $n = N-2$, let $v_1, \ldots, v_n$ be the nonzero critical values, and write*

$$\prod_{j=1}^{n}(V - v_j) = V^n + c_1 V^{n-1} + \cdots + c_n.$$

Then

$$\Lambda_N : \mathcal{A}_N^\circ \longrightarrow \mathbb{P}(1,2,\ldots,n), \qquad H \longmapsto [c_1 : \cdots : c_n]$$

is generically finite of degree

$$\boxed{\deg \Lambda_N = (N-2)N^{N-3}}.$$

If $\overline{\mathcal{A}}_N^{\mathrm{LL}}$ is the normalized graph closure in the marked admissible-cover compactification and $\eta = c_1(\mathcal{O}_{\mathbb{P}(1,\ldots,n)}(1))$, then

$$\int_{\overline{\mathcal{A}}_N^{\mathrm{LL}}} \overline{\Lambda}_N^* \eta^{N-3} = \frac{N^{N-3}}{(N-3)!}.$$

Proof. The classical polynomial LL map has degree N^{N-2}. Generic precomposition by μ_N acts freely on its monic centered source and preserves every critical value. Quotienting by affine source isomorphism therefore leaves N^{N-3} covers. Requiring one specified critical value to be zero is a transverse target slice and does not change that count. Its fiber profile is $(2, 1^{N-2})$, so marking an unramified point over zero gives $N-2$ choices. The source conditions $p_0 = 0, p_1 = 1$ and the target condition $H'(1) = -1$ then choose a unique normalized representative. This proves the degree formula.

The weighted-projective stack satisfies $\int_{\mathbb{P}(1,\ldots,n)} \eta^{n-1} = 1/n!$. Multiplying by the generic degree and using $n = N-2$ gives

$$\frac{(N-2)N^{N-3}}{(N-2)!} = \frac{N^{N-3}}{(N-3)!},$$

as asserted. ∎

Theorem 8.11 (caustic and Maxwell boundary classes)**.** *Put $n = N-2$ and work on $\overline{\mathfrak{B}}_N^{\mathrm{root}} = [\overline{M}_{0,n+2}/S_n]$. Its boundary-orbit classes satisfy the unique relation*

$$\sum_{k=1}^{n-1} k(n-k)\Delta_{0,k} = \sum_{k=2}^{n} k(k-1)\Delta_k^{\mathrm{free}}. \tag{6.17}$$

Thus

$$\mathcal{B}_n = \{\Delta_2^{\mathrm{free}}, \ldots, \Delta_n^{\mathrm{free}}, \Delta_{0,1}, \ldots, \Delta_{0,n-2}\}$$

is a basis of the rational Picard group. If $\mathcal{C}_N$ and $\mathcal{M}_N$ denote the closures of the caustic and Maxwell divisors, then in the redundant orbit presentation

$$\begin{aligned}
[\mathcal{C}_N] &= -\sum_{k=2}^{n}(k-1)(k-2)\Delta_k^{\mathrm{free}} + \sum_{k=1}^{n-1} k(n-k)\Delta_{0,k}, \\
[\mathcal{M}_N] &= -\frac{1}{2}\sum_{k=2}^{n}(k-1)(k-2)(k-3)\Delta_k^{\mathrm{free}} \\
&\quad + \frac{1}{2}\sum_{k=1}^{n-1} k(n-k)(n+k-2)\Delta_{0,k}.
\end{aligned} \tag{6.18}$$

In the basis $\mathcal{B}_n$ these become

$$[\mathcal{C}_N] = 2\sum_{k=2}^{n}(k-1)\Delta_k^{\mathrm{free}}, \tag{6.19}$$

and

$$\begin{aligned}[\mathcal{M}_N] = \frac{1}{2}\sum_{k=2}^{n}&(k-1)\big((2n-3)k-(k-2)(k-3)\big)\Delta_k^{\mathrm{free}}\\ &-\frac{1}{2}\sum_{k=1}^{n-2}k(n-k)(n-k-1)\Delta_{0,k}.\end{aligned} \tag{6.20}$$

Moreover $[\mathcal{C}_N] = 2\kappa_1$.

Proof. The logarithmic differential dH/H has residue vector $(2, 1^n, -N)$. Every proper partial sum is nonzero, so on a stable pointed source it has a nonzero residue at every node and its zero divisor avoids the nodes. If K is the relative dualizing class and D the marked divisor, the double-zero locus has class

$$\pi_*((K+D)(2K+D)) = 2\big(\sum_i \psi_i - \delta\big) = 2\kappa_1.$$

The symmetric boundary expansion of κ_1, together with the averaged Keel relation (6.17), gives (6.18)–(6.19) for the caustic.

For Maxwell, label the simple roots temporarily, normalize one of them to 1, and write $H = W^2\prod_i(W-a_i)$ and $K_H = H'/W$. On the interior,

$$\mathrm{Disc}_V\,\mathrm{Res}_W(K_H, V-H) = \mathrm{unit}\cdot\mathcal{C}_N^3\mathcal{M}_N^2.$$

A free collision of k roots gives orders $(k-1)(k-2)$, $(k-1)(k-2)(k-3)/2$, and $k(k-1)(k-2)$ for caustic, Maxwell, and the left side. If k roots tend to zero, the first two orders are $k(k-1)$ and $k(k-1)^2/2$. If $q = n-k$ roots tend to infinity, their pole orders are

$$q(q-1+2k), \qquad \frac{q}{2}\big((n-1)(q-1+2k)+k(k-1)\big).$$

Averaging over the n choices of normalized simple root gives the $\Delta_{0,k}$ coefficients in (6.18), while a free collision contributes the negative of its vanishing order. Finally (6.17) gives (6.20). The powers 3 and 2 have positive pullback orientation. On the stable target its discriminant pullback also contains target-boundary components; the pole terms above have merely been moved across the principal-divisor equality, so (6.18) does not suppress that boundary contribution. □

Theorem 8.12 (formal comparison at arbitrary root collisions)**.** *Pull the marked admissible-cover compactification of Proposition 8.7 back along the pencil $H - sW$ and mark a point over $-t$. The normalized Stein factor of this incidence is the canonical finite cover*

$$X_H = \mathrm{Spec}\,k[s,t,W]/(H(W) - sW + t),$$

and the normalized Stein factor of its critical-point incidence is $\widetilde{D}_H = \mathrm{Spec}\,k[r]$ with map (6.1). After contraction of all admissible source and target bubbles, their completed boundary charts and normalization–conductor square agree with the repository constructions, including at simultaneous collisions of arbitrarily many root clusters.

More explicitly, if the distinct multiple roots are ρ_i, of multiplicity m_i, and

$$e_i = m_i - 1, \qquad E = \sum_i e_i,$$

then the conductor in the completed normalization over $(s,t)=(0,0)$ *is*

$$\bar{\mathfrak{c}}_{(0,0)} = \bigoplus_i u_i^{e_i(E-1)} k[[u_i]]. \tag{6.21}$$

Proof. On the dense nondegenerate locus the marked-point incidence is $H(W)-sW+t=0$. This hypersurface is smooth, since its t-derivative is one, and is finite of degree N over $k[s,t]$. Its coordinate ring and the normalized Stein algebra of the proper admissible incidence are finite normal $k[s,t]$-algebras with the same generic function algebra.[6] By uniqueness of integral closure they coincide. The identical argument for the critical marking gives $k[r]$ and the map $r \mapsto (H'(r), rH'(r)-H(r))$.

At any primitive root α, put $W=\alpha+u$. Completion and elimination of t give

$$\widehat{\mathcal{O}}_{X_H,(0,0,\alpha)} = k[[s,u]], \qquad t = s(\alpha+u) - H(\alpha+u), \tag{6.22}$$

so Hensel separation makes simultaneous distinct clusters a product of the same regular charts on both sides. The reconstruction coordinates are the same rational functions in the common function field, so normality identifies their pole divisors and maximal regularity opens as well.

For a multiple root ρ_i, write $H(\rho_i+u)=c_i u^{m_i}+O(u^{m_i+1})$. On its discriminant branch, with $q_i = t-\rho_i s$, formula (6.1) gives

$$\operatorname{ord}_{u_i} s = e_i, \qquad \operatorname{ord}_{u_i} q_i = e_i+1. \tag{6.23}$$

Thus its intrinsic conductor exponent is $e_i(e_i-1)$. Distinct roots give distinct tangent lines $t=\rho_i s$, hence $I(D_i,D_j)=e_ie_j$. Applying the adjunction conductor formula (6.5a) to $f=\prod_i f_i$ adds these pairwise intersections on branch i and yields

$$e_i(e_i-1) + \sum_{j\neq i} e_ie_j = e_i(E-1),$$

which proves (6.21). Since both constructions have the same finite normalization map, quotienting by this common conductor gives the same conductor square. The argument descends from geometric branches by Galois invariance. □

Corollary 8.13 (degreewise weighted stable moduli)**.** *For every* $N\geq 4$*, weighted degree-*N *maps contain an* $(N-3)$*-dimensional family of stable polynomial left–right classes.*

Proof. By Proposition 8.2, the domain of the decorated-normalization moduli map contains a nonempty open in every degree. The map is generically finite by the preceding argument and its invariant is constant on stable classes. Its stable-class image therefore has dimension $N-3$. □

9 Proof of the degreewise theorem

Proof of Theorem 1.1. Put $n=N-1$. Every proper positive divisor r of n determines $m = n/r-1 \geq 1$, and every positive solution of $r(m+1)=n$ arises this way. There are $\tau(n)-1$ cancellation types. Lemmas 3.3–7.2 attach to type (m,r) the stable signature

$$(e_\Delta,\mu) = \left(r+1,\, n\left(\frac{n}{r}-1\right)\right). \tag{7.1}$$

[6] We use the Noetherian Stein factorization theorem in its relative-normalization form; see Stacks Project, Tag 03H0.

Distinct divisors give distinct first entries (and also distinct second entries), so cancellation maps of different types are pairwise stably inequivalent. For the type belonging to r, Lemma 3.3 gives $mr = n - r$ parameter roots, and Theorem 6.2 separates all of them. Thus the number of cancellation classes supplied by the construction is

$$\sum_{\substack{r|n \\ r<n}} (n-r) = n(\tau(n)-1) - \{\sigma(n)-n\} = n\tau(n) - \sigma(n). \tag{7.2}$$

Lemma 3.2 supplies one more degree-N map. Its boundary contact is reduced, whereas every cancellation contact is nonreduced, so Lemma 5.2 separates it from all cancellation types. The total number of classes is therefore at least

$$1 + n\tau(n) - \sigma(n) = 1 + (N-1)\tau(N-1) - \sigma(N-1).$$

Both construction lemmas give a fiber of N distinct points, completing the proof. $\square$

10 Perspective

The cubic map is the smallest marked-root example: ordering its three roots reveals the type-A_2 collision walls, and reconstruction decides which walls remain affine and which become dicritical boundary. Higher weighted maps retain simple discriminant inertia; cancellation maps replace it by ramification $r+1$ and create a thick contact with the second boundary vertex. The theorem shows that generic degree alone is a coarse invariant: its discrete stable multiplicity already has lower bound $1 + (N-1)\tau(N-1) - \sigma(N-1)$, in addition to the positive-dimensional weighted locus.

The argument deliberately stops short of an exhaustion theorem. It now distinguishes all cancellation parameter roots for fixed (m, r), as well as the different cancellation types and the weighted locus, but it does not assert that the displayed families exhaust all stable classes.